

No. of Functional Features Included in the Solution

Date	30 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the OptiCrop project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contrib.
1	Crop Recommendation	Predicts the most suitable crop based on N, P, K, temperature, humidity, pH, and rainfall inputs using ML	ML Model	Done	High
2	Data Analysis	Univariate, bivariate, and multivariate analysis of agricultural dataset using Matplotlib and Seaborn	Jupyter Notebook	Done	High
3	Data Preprocessing	Null value handling, outlier removal using IQR, seasonal crop extraction, train/test split	Jupyter Notebook	Done	High
4	K-Means Clustering	Groups similar agricultural conditions using Elbow method to find optimal clusters	ML Model	Done	Medium
5	Logistic Regression	Classifies and predicts the best crop with 94% accuracy using trained model	ML Model	Done	High
6	Flask Web Application	Responsive web app with Home, About, and FindYourCrop pages for user interaction	Flask App	Done	High
7	Model Persistence	Trained ML model saved as model.pkl using Pickle for fast reuse without retraining	app.py	Done	High

8	Real-time Prediction	User enters parameters on web form → Flask backend processes → returns instant crop recommendation	Flask + ML	Done	High
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Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8

Core / Must-Have Features	6 (Crop Recommendation, Data Analysis, Preprocessing, Logistic Regression, Flask App, Real-time Prediction)
Additional / Nice-to-Have Features	2 (K-Means Clustering, Model Persistence)
Features Tested & Verified	8 (All features tested locally and verified)

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Home page, About page, FindYourCrop prediction form
2	Backend / Logic	3	Flask routes, Prediction endpoint, Input validation
3	Database / Storage	2	CSV dataset storage, Pickle model serialization
4	ML / AI	3	K-Means Clustering, Logistic Regression, Real-time prediction
5	Security / Authentication	1	Input validation, warnings suppression, error handling